

ML learnings using open source tools like Spark, Scikit-learn or TensorFlow. The data is then exported to MLeapBundle, which can be deployed at the edge data centre. This approach reduces any network latency concerns typical data centre connectivity will have, allowing for enterprise grade processing power for real-time machine learning.

The final step in request processing is to interface with core and third party systems of the organisation for the data necessary in experience analytics. This data runs in a typical data centre or in cloud native services. The end-to-end low latency application can be developed using proven open source tools.

The technical architecture shown in Figure 2 is a reference implementation of a low latency application that could be developed for an interactive experience for various common use cases. Each of the open source tools selected has many alternatives as well. Depending on the capability and functional need of the solution, the appropriate tools may be swapped to build the right-sized implementation. Links to each of the tools used, with a

brief on their purpose, are given in the *References* at the end of this article.

Most popular cloud service providers are embracing advancements in low latency solutions. They either offer a cloud native offering that's wrapped around familiar open source tools (e.g., Apache Spark) or they build their own, providing developers the opportunity to select the right tools that are cost-

effective, but meet the expectations of low latency and good performance.

Deploying a scalable low latency solution that meets the needs of an organisation can enable it to deliver better products and services to its customers. Though there is considerable effort involved in deploying scalable low latency solutions the right way, the investment and risks are worth it. **END** 

References

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- Memcached - Distributed memory object caching system, <https://memcached.org/>
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- TensorFlow - Machine learning platform, <https://www.tensorflow.org/>

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4. FaaS provides you with the ability to be agile, i.e., to write functional code.

Blockchain Platform as a Service (BPaaS)

Blockchain has taken the tech industry by storm in recent years. It is one of the most in-demand technologies right now, surpassed marginally by AI and data science related technologies. What makes blockchain so attractive is its open-ledger architecture providing security, scalability, and transparency. These features are necessary for many applications such as banking, electoral systems, and even social media. With such wide-ranging applications, it becomes necessary to be able to host such products on the cloud

with a model that specifically caters to the needs of this technology. This is where BPaaS comes into the picture. Many companies today, including big names such as Amazon AWS and Microsoft Azure, are providing BPaaS solutions for customers specifically looking to host blockchain based apps in the cloud.

Benefits of BPaaS

1. It caters to the specific needs of the blockchain industry such as support

for custom languages used for writing smart contracts.

2. Supports integrations with pre-eminent blockchains such as Ethereum by providing API bridges.
3. Supports custom databases used in the application life cycle of blockchain technologies.
4. It has all the goodness of the cloud with the 'pay as you go' feature, scalability, security, and ease of access. **END** 

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